MARSEL BAKASHOV

Software Engineering Student
Data Science / ML / RL / AI

Portfolio
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ABOUT ME

Software engineering student specializing in machine learning and reinforcement learning, with experience applying ML to large-scale systems and real-world industrial problems (insurance pricing, predictive maintenance, semantic search). Passionate about building data-driven systems that bridge research and production.

SKILLS

Programming:	Python, C++, Java, SQL, Bash, R
ML / Data:	PyTorch, Scikit-learn, TensorFlow, HuggingFace, MLflow
Tools:	Docker, Git, Airflow, Azure, Power BI, Jenkins, Databricks, Snowflake
Languages:	English, French, Russian, Spanish, Korean

EDUCATION

1/2026 – 2/2027 — **Professional Master's in Machine Learning** — ***University of Montreal / McGill - MILA***
Present — Montreal, Canada

9/2021 – 12/2025 — **Bachelor of Software Engineering** — ***Polytechnique Montreal***
GPA: 3.4 / 4.0 — Montreal, Canada

2/2025 – 6/2025 — **Bachelor of Computer Engineering** — ***Korea University College of Engineering***
Academic Exchange — Seoul, South Korea

EXPERIENCE

1/2026 – 5/2026 — **Data Scientist & RL Intern** — ***Intact Insurance***
- Developed constrained RL pricing policies using adaptive primal–dual methods, validated through simulation and risk metrics, increasing long-term profitability while lowering volatility.

`Python / PyTorch / RL / CVar / MLflow / Databricks / Snowflake / Risk Metrics`

6/2025 – 9/2025 — **Data Scientist & ML Intern** — ***Pratt & Whitney***
- Developed predictive models to estimate engine remaining useful life, enabling better maintenance planning and reducing unplanned downtime by 15%.

`Python / Scikit-learn / PyTorch / Pandas / NumPy / MLflow / Jupyter`

5/2024 – 9/2024 — **Data Scientist Intern** — ***Radio-Canada / CBC***
- Designed an AI-powered unified indexing system centralizing internal content sources for semantic and keyword-based document search (15 M + elements and used by 100+ employees).

`ELK / ETL / Airflow / Python / Azure AI Search / Hugging Face / Embeddings`

9/2023 – 12/2023 — **Full-Stack Developer Intern** — ***Stingray***
- Applied analytics to drive improvements in an Agile setup with CI, testing, and code reviews.

`Next.js / React / Node.js / TypeScript`

PROJECTS

Personal Project — **GravityGuyML — Reinforcement Learning Side-Scroller** — ***GitHub***
- Built a Gymnasium-compatible Pygame environment with procedural generation and deep RL agents (`DQN`, `PPO`, `SAC`, `A2C`).

`Python / Machine Learning / Deep Learning / Reinforcement Learning / TensorFlow`

Personal Project — **Quantitative Finance Notes — Monte Carlo & GBM Modeling** — ***GitHub***
- Built reproducible notebooks on stochastic processes, Monte Carlo simulations, and Geometric Brownian Motion for market modeling and data calibration.

`Python / NumPy / Pandas / Matplotlib / Monte Carlo / Quantitative Finance / GBM`

Academic Project — **Weather-Based Disease Outbreak Prediction** — ***GitHub***
- Predicted outbreaks from climate data using classification models (Logistic Regression, SVM, Decision Trees) with F1-score and AUC-ROC evaluation.

`Python / Jupyter / pandas / scikit-learn / Matplotlib / FP-Growth`

EXTRACURRICULAR ACTIVITIES

1/2024 – 8/2024 — **Teaching Assistant – Discrete Mathematics** — ***Polytechnique Montreal***
- Led review sessions, graded assignments, and supported 80+ students.

1/2024 – 5/2024 — **Sponsorship Director** — ***PolyAI Hackathon***
- Raised over $22,000 in funding and led a sponsorship team with weekly meetings.